SHORT COARSENINGS DETERMINE RIBBON SCHUR EQUIVALENCE

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Abstract. A ribbon Schur equivalence class is determined by coarsening coefficients of logarithmic block depth, even when those coefficients are recorded modulo three. More precisely, unequal coarsening inventories have a distinguishing coefficient of magnitude one or two within an explicit bound depending on the divisors of the weight. The proof gives an alternative between a short distinguishing coarsening and, after reversal if needed, a literal common inner factor. A counting identity detects failures of periodicity; reflection produces positive block lengths whose sum is the weight of the given compositions. Integral substitution permits descent, yielding an independent proof of the factor-reversal classification and compatibility with composition on both sides, including arbitrary repetitions with adjacent copies joined by merging their meeting parts. Two recursive families determine the exact cumulative distinguishing depth. A parity example shows that the arithmetic cutoff does not extend unchanged to coefficients modulo two.

1. Introduction

Different ribbons can have the same Schur function. A ribbon is encoded by the composition of its row lengths, read from bottom to top. A composition is an ordered list of positive integers. Merging adjacent parts and sorting the resulting sums produces a partition; recording all such partitions with multiplicity gives the composition's *coarsening inventory*. For example, $(1,2,1)$ gives (4), two copies of $(3,1)$, and $(2,1,1)$. Multiplicity matters: equality of sets of outcomes is weaker than equality of inventories.

Equality of coarsening inventories is equivalent to equality of the corresponding ribbon Schur functions [1, Theorem 2.6]. Billera, Thomas, and van Willigenburg introduced an operation $\circ$ on compositions and described the equivalence classes by reversal of factors in a $\circ$-factorization [1, Theorem 4.1]. The present paper gives an independent quantitative reconstruction argument. Shallow coarsening data force the factor description. For a fixed inner composition, composition acts by substitution on inventories. Replacing that inner composition by an equivalent one is a different assertion: its inventory does not directly specify the inventories of its near-powers (repeated copies joined by merging the meeting parts). The reconstruction in subsequent sections proves that these substitutions agree, rather than assuming it during factor extraction.

1.1. Main results.

A logarithmic block-depth truncation of the coarsening inventory determines the entire ribbon Schur equivalence class.

For a composition β of weight n, let $\ell(\beta)$ be its number of parts and β^* its reversal. Write $\mathrm{sort}(\rho)$ for the partition obtained by sorting a list ρ in weakly decreasing order, and $h_\lambda = \prod_i h_{\lambda_i}$ in the independent complete homogeneous generators h_j. Define

$$G_\beta = \sum_{\rho \succeq \beta} h_{\mathrm{sort}(\rho)}, \qquad m_\beta(\lambda) = [h_\lambda] G_\beta, \tag{1.1}$$

where ρ ranges over coarsenings. Write $\mathrm{Comp}(n)$ for the set of compositions of n, and $\beta \sim \gamma$ when $G_\beta = G_\gamma$. Over $\mathbb{Z}$, the weight is $\mathrm{wt}(h_j) = j$, whereas the ordinary degree is $\deg_h(h_j) = 1$.

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The latter counts blocks. A superscript $[\le r]$ denotes truncation in ordinary degree. Let $\Omega(n)$ count prime factors with multiplicity, with $\Omega(1) = 0$.

Let $B(n)$ denote the integer cutoff defined in Definition 7.3.

Theorem 1.1 (Small discrepancies)**.** *For compositions β, γ of weight n, either their inventories are equal or there is a partition $\lambda \vdash n$ such that*

$$\ell(\lambda) \le B(n) \le \Omega(n) + 2, \qquad m_\beta(\lambda) - m_\gamma(\lambda) \in \{-2, -1, 1, 2\}.$$

Here $B(n)$ is the explicit arithmetic bound in Definition 7.3; by Proposition 7.5, it satisfies $B(n) \le 2\log_6 n + \log_6 9$. The partition can be chosen with all parts distinct except possibly one repeated pair. For every integer $q \ge 3$,

$$G_\beta^{[\le B(n)]} \equiv G_\gamma^{[\le B(n)]} \pmod q \quad \Longleftrightarrow \quad G_\beta = G_\gamma. \tag{1.2}$$

The logarithm bounds the number of blocks in a distinguishing coarsening, not the number of coefficients supplied or the running time of a recognition algorithm. The modular assertion is proved only after the relevant integer multiplicities have been computed and bounded.

Write $\cdot$ for concatenation and $\odot$ for concatenation with the two meeting parts merged. The near-power $\epsilon^{\odot j}$ is the near-concatenation of j copies of ϵ. The structural output uses the homomorphism

$$\Psi_\epsilon : \mathbb{Z}[h_1, h_2, \ldots] \longrightarrow \mathbb{Z}[h_1, h_2, \ldots], \qquad \Psi_\epsilon(h_j) = G_{\epsilon^{\odot j}}. \tag{1.3}$$

The composition product is $\alpha \circ \epsilon = \epsilon^{\odot\alpha_1} \cdot \cdots \cdot \epsilon^{\odot\alpha_{\ell(\alpha)}}$.

Theorem 1.2 (Structure and composition)**.** *For compositions β, γ, the following are equivalent:*

1. *$G_\beta = G_\gamma$;*
2. *$\Psi_\beta = \Psi_\gamma$;*
3. *there are factorizations $\beta = \delta_1 \circ \cdots \circ \delta_s$ and $\gamma = \epsilon_1 \circ \cdots \circ \epsilon_s$ with $\epsilon_i \in \{\delta_i, \delta_i^*\}$ for every i.*

Consequently, inventory equivalence is a two-sided congruence for $\circ$. In particular,

$$G_\beta = G_\gamma \quad \Longrightarrow \quad G_{\beta^{\odot m}} = G_{\gamma^{\odot m}} \qquad (m \ge 1). \tag{1.4}$$

The factor-reversal characterization in this theorem is the classification of [1], reproved here as an output of the quantitative descent. The implication from inventory equality to equality of the maps is substantive; its converse is evaluation at h_1. This concerns the particular family $\{\Psi_\beta\}$, not arbitrary polynomial endomorphisms.

For unequal inventories of equal weight define

$$\operatorname{dep}(\beta, \gamma) = \min\{\ell(\lambda) : m_\beta(\lambda) \ne m_\gamma(\lambda)\},$$

and put

$$D_=(n) = \max_{\substack{\beta,\gamma \in \operatorname{Comp}(n) \\ G_\beta \ne G_\gamma}} \operatorname{dep}(\beta, \gamma), \qquad C(N) = \max_{1 \le n \le N} D_=(n). \tag{1.5}$$

An empty maximum is zero. The fixed-weight optimum $D_=(n)$, the cumulative optimum $C(N)$, and the sufficient cutoff $B(n)$ are different quantities.

Theorem 1.3 (Exact cumulative depth)**.** *For every positive integer N, the maximum first distinguishing depth among weights at most N is*

$$C(N) = \begin{cases} 0, & N = 1, \\ 2, & 2 \le N \le 4, \\ 3, & 5 \le N \le 11, \\ 2j + 4 + \mathbf{1}_{\{N \ge 30 \cdot 6^j\}}, & 12 \cdot 6^j \le N < 72 \cdot 6^j, \quad j \ge 0. \end{cases} \tag{1.6}$$

Both threshold families are attained by recursively defined pairs with a unit first discrepancy. The same cumulative frontier is sharp for the modular tests in (1.2)*, for every $q \ge 3$.*

The lower-bound construction multiplies weight by six and increases first distinguishing degree by two. Its induction hypothesis is deliberately stronger than equality for a single pair: it concerns every near-concatenation of an odd number of alternating factors. No recognition theorem or near-power compatibility is used in that construction.

1.2. **Original-weight detection and near-power stability.** Two- and three-block counts give odd functions with disjoint support satisfying a triangle identity. Elementary support separation confines the changed cuts to a proper cyclic grid. A chain of masked periods then isolates the remaining neutral positions. At each failed period, antichain cancellation produces a marked coefficient difference which is a product of local binary differences and a terminal three-probe expression. A reflected packing argument makes all its parts positive with total *exactly the original weight* n of the given compositions. If no short discrepancy exists, the chain stabilizes at a literal common inner factor. Grid projection removes that factor, and unsigned substitution puts it back.

The original-weight argument also supplies an implication used in the proof of Billera, Thomas, and van Willigenburg [1]. With $\mathcal{E}(\beta) = \{\gamma : G_\gamma = G_\beta\}$, the specialization of [1, Theorem 4.5] to a one-part outer composition is

$$\mathcal{E}((m) \circ \beta) \subseteq \{(m) \circ \gamma : \gamma \in \mathcal{E}(\beta)\}. \tag{1.7}$$

We prove this inclusion directly in Corollary 6.4; the reverse containment is (1.4). The proof of [1, Lemma 5.1] uses equality of these sets, citing that theorem. The cited theorem alone does not supply the reverse containment. Invoking the completed classification would be circular in that deduction: its printed proof uses Lemma 5.1 in the enlargement steps of Lemmas 5.9 and 5.10. Our argument makes no such enlargement. The issue is this missing implication in the specified deduction, not falsity of the classification statement.

The present results concern ribbons. Ribbon-related composition operations also enter constructions of more general skew Schur equalities [2, 3], but no classification for arbitrary skew diagrams is asserted here. Extension would require new analogues of support separation, literal factor extraction, and positive detection inside the original diagram.

1.3. **Organization.** Section 2 develops coarsening inventories, the integer multiplicity convention, and unsigned composition as substitution. Sections 3 to 5 establish low-order balance and support separation, develop period masks and marked coarsenings, and realize the resulting coefficient differences at the original weight. Section 6 completes descent and proves composition congruence; Section 7 optimizes the detector budget and proves matching lower bounds. The lower-bound construction uses only the seam identities from Section 2, independently of recognition and composition compatibility. The near-power class identity (Corollary 6.4) in Section 6 is then proved directly from the local alternative and (1.4). Sections 8 and 8.5 give factor recovery, effective coefficient checking, and the characteristic-two limitation. The support proof is included in Section 3; the arithmetic optimization is proved in Section 7. Appendix A gives a multilevel example.

2. Coarsening inventories and unsigned composition

2.1. **Cuts, gradings, and multiplicities.** A composition $\beta = (\beta_1, \dots, \beta_k)$ has weight $|\beta| = \sum_i \beta_i$, length $\ell(\beta) = k$, reversal β^*, and cut set

$$S(\beta) = \{\beta_1, \beta_1 + \beta_2, \dots, \beta_1 + \cdots + \beta_{k-1}\}.$$

The empty composition is used only for empty prefixes and suffixes, with $G_\varnothing = 1$. The polynomial ring $\Lambda_{\mathbb{Z}} = \mathbb{Z}[h_1, h_2, \dots]$ can be identified with the ring of symmetric functions in the complete homogeneous generators. Throughout, $h_0 = 1$. For homogeneous components, write $F^{[r]}$ for ordinary degree and $[F]_{\mathrm{wt}=s}$ for weight.

For a labeled positive list $\mathbf{a} = (a_1, \dots, a_r)$ of total n, and the cyclic indicator b of $S(\beta)$ with $b(0) = 0$, put

$$K_b(\mathbf{a}) = \sum_{\pi \in \mathfrak{S}_r} \prod_{j=1}^{r-1} b(a_{\pi(1)} + \cdots + a_{\pi(j)}). \tag{2.1}$$

Labels distinguish positions even when their numerical values coincide. For $\lambda = \mathrm{sort}(\mathbf{a})$, set $\kappa(\lambda) = \prod_j \nu_j(\lambda)!$, where ν_j is the multiplicity of the part j.

For a second composition γ of weight n with cut indicator c, write $\Delta K(\mathbf{a}) = K_b(\mathbf{a}) - K_c(\mathbf{a})$ and $\Delta m(\lambda) = m_\beta(\lambda) - m_\gamma(\lambda)$. The arguments are omitted when fixed; ΔK and Δm are difference notation, not products.

Lemma 2.1. *Over $\mathbb{Z}$,*

$$K_b(\mathbf{a}) = \kappa(\lambda) m_\beta(\lambda), \qquad \Delta K = \kappa(\lambda) \Delta m. \tag{2.2}$$

If $0 < |\Delta K| \le 2$, then $\Delta m \in \{-2, -1, 1, 2\}$ and λ has distinct parts except possibly one repeated pair.

Proof. Every numerical ordering of λ is produced by exactly $\prod_j \nu_j(\lambda)!$ permutations of its labels. It contributes if and only if all proper partial sums are retained cuts. The final assertions follow by exact integer divisibility: a positive product of factorials not exceeding two is either one or two. □

Remark 2.2. Path cancellation takes place over $\mathbb{Z}$ throughout. The bound on ΔK is first converted by (2.2), before Δm is reduced modulo $q \ge 3$. There is no division by $\kappa(\lambda)$ in a finite field. For instance, $K_b(1,1,1) = 6$ for $\beta = (1,1,1)$, while the ordinary coefficient is one. Numerical coincidences among reflected or padded probes are allowed; their multiplicities are recalculated after the positive list has been formed.

The standard ribbon expansion [1, Proposition 2.1] is

$$r_\beta = (-1)^{\ell(\beta)} \mathcal{A}(G_\beta), \qquad \mathcal{A}(h_j) = -h_j. \tag{2.3}$$

Since the top ordinary degree of G_β is $\ell(\beta)$, equality of ribbon functions implies equal lengths and then equal inventories; the converse is immediate. This is [1, Theorem 2.6].

2.2. **Composition as substitution.** Concatenation is denoted $\cdot$. Near-concatenation $\alpha \odot \beta$ merges the two meeting parts, and

$$\alpha \circ \epsilon = \epsilon^{\odot \alpha_1} \cdot \cdots \cdot \epsilon^{\odot \alpha_{\ell(\alpha)}}.$$

If $p = |\epsilon|$, then

$$S(\alpha \circ \epsilon) = pS(\alpha) \ \cup \bigcup_{j=0}^{|\alpha|-1} \big(jp + S(\epsilon)\big). \tag{2.4}$$

This proves associativity and $(\alpha \circ \epsilon)^* = \alpha^* \circ \epsilon^*$ directly. In particular,

$$S(\beta^{\odot m}) = \bigcup_{j=0}^{m-1} (jn + S(\beta));$$

there is no cut at an internal seam jn.

Lemma 2.3. *A composition β of weight n has a literal inner factor of weight $p \mid n$ precisely when its cuts off the p-grid repeat with period p. At that weight the factors are unique: the inner cuts are $S(\beta) \cap \{1, \dots, p-1\}$, and the outer cuts are $\{j : jp \in S(\beta)\}$. Two compositions have the same literal inner factor when their off-grid cuts agree and repeat with that period.*

Proof. The cut formula (2.4) proves necessity and uniqueness. Conversely, repeat the first open p-cell and retain exactly the specified grid cuts. Their union is the original cut set, so that formula gives the claimed factorization. □

Proposition 2.4. *For all compositions,*

$$G_{\alpha\cdot\beta} = G_\alpha G_\beta + G_{\alpha\odot\beta}, \tag{2.5}$$

$$\Psi_\epsilon(G_\alpha) = G_{\alpha\circ\epsilon}, \tag{2.6}$$

$$\Psi_{\alpha\circ\epsilon} = \Psi_\epsilon\Psi_\alpha, \qquad \Psi_{\epsilon^*} = \Psi_\epsilon. \tag{2.7}$$

For

$$\pi_p(h_j) = \begin{cases} h_{j/p}, & p \mid j, \\ 0, & p \nmid j, \end{cases} \qquad p = |\epsilon|, \tag{2.8}$$

one has

$$\pi_p\Psi_\epsilon = \mathrm{id}. \tag{2.9}$$

All these are integral identities and remain valid over any commutative coefficient ring.

Proof. Retaining or erasing the joining seam proves (2.5). For (2.6), group coarsenings by the retained outer seams. An outer coarsening $\rho \succeq \alpha$ leaves independently coarsened intervals $\epsilon^{\odot\rho_i}$; their contribution is $\prod_i G_{\epsilon^{\odot\rho_i}}$.

The cut identity gives $(\alpha\circ\epsilon)^{\odot j} = \alpha^{\odot j}\circ\epsilon$, proving the composition law on every generator. Literal reversal preserves an inventory and commutes with near-powers, proving the second identity in (2.7).

A coarsening of $\epsilon^{\odot j}$ with all blocks divisible by p can retain only p-grid cuts. There are none, so its only such coarsening is (pj). Hence $\pi_p(G_{\epsilon^{\odot j}}) = h_j$, proving (2.9). □

Corollary 2.5. *Reversing any specified factors of a composition factorization preserves G. This assertion uses neither recognition nor* (1.4).

Proof. Apply the identities (2.7) successively and then evaluate on h_1. Literal reversal of an outer composition also preserves its inventory. □

Remark 2.6. The compatibility theorem concerns individual inventories, not a linear extension of near-squaring on their span. Indeed,

$$G_{(1,1,4)} + G_{(1,2,3)} - G_{(1,4,1)} - G_{(2,1,3)} = 0,$$

whereas the same combination of their near-square inventories has coefficient one at $h_4h_3^2h_2$. Thus no $\mathbb{Q}$-linear map on the span of weight-six inventories sends every G_α to $G_{\alpha^{\odot 2}}$. This does not conflict with Ψ_ϵ: its inner composition is fixed. Reconstruction first determines a class nonlinearly and then proves compatibility on that class.

3. Low-order balance and support separation

3.1. **The triangle identity.** Suppose first that the two-block inventories of $\beta, \gamma \in \mathrm{Comp}(n)$ are equal. Write b, c for their cyclic cut indicators and set

$$\tau(x) = b(x) + b(-x) = c(x) + c(-x), \qquad u(x) = b(x) - c(-x), \qquad v(x) = b(x) - c(x). \tag{3.1}$$

Both u, v are odd, take values in $\{-1, 0, 1\}$, and have disjoint supports:

$$u(x)v(x) = 0.$$

At a midpoint, $\tau(n/2) = 2b(n/2)$; the ordinary two-block coefficient there is $b(n/2)$, not $\tau(n/2)$.

For positive a, b', c' summing to n, put $x(t) = b(t) - b(-t)$. Pairing opposite orders of the three labeled parts gives

$$K_b(a, b', c') = \frac{1}{2}\sum_{\{i,j\}\subseteq\{1,2,3\}} \big(\tau(a_i)\tau(a_j) - x(a_i)x(a_j)\big), \quad (a_1, a_2, a_3) = (a, b', c').$$

For the two originals, these odd functions are $u + v$ and $u - v$. Consequently

$$K_b(\mathbf{a}) - K_c(\mathbf{a}) = -\sum_{i<j}\big(u(a_i)v(a_j) + v(a_i)u(a_j)\big). \tag{3.2}$$

Lemma 3.1. *If there is no ordinary coefficient discrepancy of magnitude one or two through three blocks, then the two-block data agree and*

$$u(x)v(y)+v(x)u(y)=u(x+y)\big(v(x)+v(y)\big)+v(x+y)\big(u(x)+u(y)\big) \tag{3.3}$$

holds on $\mathbb{Z}/n\mathbb{Z}$. Conversely, after two-block equality, equality through three blocks is equivalent to (3.3).

Proof. Ordinary two-block coefficients lie in $\{0,1,2\}$, so any nonzero difference already has magnitude one or two. This proves (3.1).

For any three positions with disjoint ternary pairs (u_i,v_i), at most two of the cross-products in (3.2) are nonzero. Its absolute value is therefore at most two. By Lemma 2.1, a nonzero value supplies the stated ordinary certificate.

Taking $a_3=-a_1-a_2$ converts (3.2) into (3.3). Positive representatives with $a_1+a_2<n$ give all such triangles. If $a_1+a_2>n$, negate both arguments; the products in (3.3) are unchanged. Zero arguments and opposite arguments give identities directly from oddness and disjointness. This proves the full cyclic statement and its converse. □

3.2. **Elementary support separation.** All matrices here have at least two vertices. A skew-ternary matrix is a skew-symmetric matrix with entries in $\{-1,0,1\}$. A nontrivial partition $X=P\sqcup Q$ is a *constant cut* if all entries A_{pq}, $p\in P$, $q\in Q$, are equal. Call A inseparable if it has no constant cut. A subset C is a *clan* if every vertex outside C has the same entry to every vertex of C.

Lemma 3.2. *Every inseparable skew-ternary matrix has an inseparable principal submatrix on three or four vertices.*

Proof. Suppose every three- and four-vertex principal submatrix is separable. Define $x<y$ when $A_{xy}=1$. If $x<y<z$, then $A_{xz}=1$: a zero would give an inseparable directed two-edge path, and -1 an inseparable directed triangle. Hence $<$ is a strict partial order.

Let H be its undirected comparability graph. An induced four-vertex path would give an inseparable four-vertex matrix. Indeed, the path and its complement are both connected, excluding respectively a zero constant cut and a nonzero constant cut. Thus H has no induced four-vertex path.

A connected graph on at least two vertices with no induced four-vertex path has disconnected complement. Induct on its number of vertices and remove z. If the remaining graph is disconnected, z must be adjacent to every remaining vertex. Otherwise, in a component containing both a neighbor and a nonneighbor of z, take an edge crossing these types; a neighbor of z in a different component completes an induced four-vertex path. Connectivity ensures that z has a neighbor in every component. Thus the complement has isolated vertex z.

If the remaining graph is connected, its complement has components $C_1,\dots,C_t$ by induction. All graph edges between different components are present. If z is mixed on C_i, a complement edge crossing its two adjacency types, together with any nonneighbor of z in another component, gives an induced four-vertex path. Hence z is adjacent to every other component, one of which remains separated in the complement. If z is uniform on every component, connectivity gives at least one component to which it is completely adjacent; that component again remains separated. This proves the graph fact.

If H is disconnected, its components give a zero constant cut in A. Otherwise the complement components are pairwise completely comparable. Along an incomparability path, the direction of comparison to an outside vertex cannot change, by transitivity. Thus the complement components are totally ordered, with uniform directions between them. An extreme component gives a nonzero constant cut. This contradicts inseparability and proves the lemma. Matrices on two vertices are themselves separable. □

Lemma 3.3. *Contracting a proper nonempty clan to one representative preserves inseparability in both directions. Every inseparable skew-ternary matrix on at least five vertices has an inseparable one-vertex deletion.*

Proof. A constant cut in the contraction expands to one in the original by the clan property. Conversely, restrict a constant cut of the original to the contraction. If both sides remain, it is still constant. If one side disappears, it lay entirely in the clan; the original constant cut and the clan property then make the representative a constant cut from all outside vertices. This proves the first assertion.

Choose an inclusion-maximal proper inseparable principal submatrix W containing a core from Lemma 3.2. If $|W| = |X| - 1$, it is the required deletion. Otherwise $W \cup \{z\}$ is separable for every $z \notin W$. Its constant cut cannot split W, so it separates z from all of W. Hence W is a clan. Delete any nonrepresentative vertex of W. The remaining clan contracts to the same inseparable contraction as before; the first assertion proves the deletion is inseparable. □

Let U, V be skew-ternary matrices on X with disjoint supports and $A = U + V$. Color their nonzero entries blue and red. Assume that every triple satisfies

$$\sum_{\mathrm{cyc}(i,j,k)} \big(U_{ij}V_{jk} + V_{ij}U_{jk}\big) = 0. \tag{3.4}$$

Lemma 3.4. *If A is inseparable, at least one of its two color graphs is disconnected.*

Proof. Every connected component of either color is a clan. For example, if xy is red and z lies outside its red component, then the zx, zy entries have no red part. Equation (3.4) gives $A_{zx} = A_{zy}$. Propagate along a red path.

For at most three vertices the two disjoint color graphs cannot both be connected. On four vertices, connectedness of both would make each a three-edge tree. Neither can be a star, whose complement is disconnected; so relabel the red graph as the path $12, 23, 34$. The triangle equations force

$$A_{12} = a, \quad A_{23} = -a, \quad A_{34} = a, \qquad A_{13} = A_{14} = A_{24} = b,$$

where $a, b \in \{1, -1\}$. If $b = a$, the cut $\{1, 3\} \mid \{2, 4\}$ is constant. If $b = -a$, the cut $\{1, 2\} \mid \{3, 4\}$ is constant. Both contradict inseparability.

For at least five vertices, choose an inseparable deletion $W = X \setminus \{z\}$ by Lemma 3.3. By induction one color, say red, is disconnected on W, with components $C_1, \ldots, C_t$. If the full red graph is connected, z has a red neighbor in every C_i. If the full blue graph is connected, z has a blue neighbor x, say in C_1. For $i > 1$, choose a red neighbor $y_i \in C_i$ of z. The edge xy_i is not red, since it joins different red components. The triangle equation forces it to be blue and gives $A_{xy_i} = A_{xz}$. Because all red components are clans in $A[W]$, every entry from C_1 to $W \setminus C_1$ equals A_{xz}. This is a constant cut in $A[W]$, a contradiction. □

Theorem 3.5 (Support separation)**.** *Let $u, v : \mathbb{Z}/n\mathbb{Z} \to \{-1, 0, 1\}$ be nonzero odd functions with $uv = 0$ satisfying* (3.3)*. At least one of their supports generates a proper subgroup of $\mathbb{Z}/n\mathbb{Z}$.*

Proof of Theorem 3.5. Define the cyclic matrices $U_{ij} = u(j - i)$ and $V_{ij} = v(j - i)$. Equation (3.3) is exactly (3.4). Suppose both supports generated the cyclic group. Both Cayley color graphs would then be connected.

Every row of $A = U + V$ sums to zero, by oddness. A constant cut $P \mid Q$ of value c would give $0 = |P||Q|c$ on summing the rows over P, since the internal skew-symmetric entries cancel. Thus $c = 0$, which would disconnect the union of the two color graphs. Therefore A is inseparable, contradicting Lemma 3.4. □

Remark 3.6. More generally, let U, V be disjoint-support skew-ternary matrices on a finite set X with at least two vertices, satisfying (3.4). If $A = U + V$ has a strictly positive real null vector π, then their support graphs cannot both be connected. Indeed, for a constant cut $X = P \sqcup Q$ of value c, skew-symmetry gives

$$0 = \sum_{i \in P} \pi_i (A\pi)_i = c \left(\sum_{i \in P} \pi_i \right) \left(\sum_{j \in Q} \pi_j \right).$$

Both sums are positive, so $c = 0$. Disjointness of the supports then forces every cross-cut entry of both matrices to vanish. If both support graphs were connected, no constant cut could exist; A would be inseparable, contradicting Lemma 3.4. The cyclic application above is the case $\pi \equiv 1$.

Proposition 3.7. *For the cut functions in* (3.1)*, assume the hypotheses of Theorem 3.5. The subgroups generated by* $\operatorname{supp} u$ *and* $\operatorname{supp} v$ *are strictly nested. After replacing* γ *by* γ^* *if necessary,*

$$g_u := \gcd(n, \operatorname{supp} u) \text{ properly divides } d := \gcd(n, \operatorname{supp} v). \tag{3.5}$$

The function

$$w(x) = \begin{cases} u(x), & d \nmid x, \\ 0, & d \mid x \end{cases} \tag{3.6}$$

is nonzero, odd and d*-periodic. Its least positive period* e *satisfies*

$$3 \le e \mid d < n, \qquad n/d \ge 3. \tag{3.7}$$

For every $e \mid p \mid d$*, the mask*

$$\mathcal{B}_p(x) = \begin{cases} b(x), & p \nmid x, \\ 0, & p \mid x \end{cases} \tag{3.8}$$

has odd part

$$\mathcal{B}_p(x) - \mathcal{B}_p(-x) = w(x). \tag{3.9}$$

Proof. Let $H = \langle \operatorname{supp} v \rangle$. If $x \in \operatorname{supp} u$ lies outside H, then for every $a \in \operatorname{supp} v$, equation (3.3) gives $u(x+a) = u(x) \neq 0$. The same holds on the entire coset $x+H$. Hence $H \subseteq \langle \operatorname{supp} u \rangle$. If no such x exists, the reverse containment holds. Equality of the subgroups is impossible: restrict the functions to that common subgroup and apply Theorem 3.5, since both restricted supports would generate it.

Reversing the second original exchanges u, v, proving (3.5). For $a \in \operatorname{supp} v$ and $d \nmid x$, (3.3) gives $u(x + a) = u(x)$. These shifts generate $d\mathbb{Z}/n\mathbb{Z}$, so (3.6) is d-periodic. It cannot vanish identically, since then d would divide g_u. An odd function of period one or two is zero; this proves $e \ge 3$. Likewise a nonzero odd function supported on the d-grid requires that grid to have order at least three.

Off the p-grid one is also off the d-grid. There $v = 0$, so $b(x) - b(-x) = u(x) = w(x)$. On the p-grid, $w = 0$ because $e \mid p$. This proves (3.9). □

If $v = 0$, the originals are identical. If $u = 0$, they are literal reversals. These two cases are excluded whenever the normalization is used. Neither equal lengths nor a factorization is assumed.

4. Period masks and marked coarsenings

Throughout this section and Section 5, retain the normalized cut functions b, c, comparison functions u, v, w, and divisors e, d from Proposition 3.7. A support element used as a block length is represented by an integer between 1 and $n - 1$.

A *detector* is a labeled coarsening-count difference ΔK for which the argument establishes $|\Delta K| \le 2$; a nonzero value gives an ordinary coefficient discrepancy by Lemma 2.1. The marked identities in Lemma 4.2 and Theorem 4.4 construct these differences. A positive detector means that its block lengths are positive, not that the difference is positive.

4.1. **The first periodic mask.**

The least period of a function on a finite cyclic group is the least positive divisor of its cyclic order that is a period.

Lemma 4.1. *Let* N_0 *be a positive integer and let* w, f *be rational-valued functions on* $\mathbb{Z}/N_0\mathbb{Z}$*. Suppose* w *is nonzero and odd, has least period* e*, satisfies* $wf = 0$*, and satisfies* (3.3) *with* w, f *in place of* u, v*. Then*

$$f(t + e) = f(t) \qquad (e \nmid t). \tag{4.1}$$

If also $f = 0$ on $e\mathbb{Z}$ and the sum of f on every residue class modulo e is zero, then $f = 0$.

Proof. Suppose $f(t+e) \neq f(t)$. For $w(a) \neq 0$, compare the equations at $(a, t-a)$ and $(a+e, t-a)$. The values of w are unchanged and $f(a) = f(a+e) = 0$. Subtracting yields

$$\big(w(a) + w(t-a)\big)\big(f(t+e) - f(t)\big) = 0.$$

Oddness therefore gives $w(a-t) = w(a)$. Translation by $-t$ maps the finite support of w into itself and preserves every nonzero value, hence preserves the whole function. Thus $e \mid t$, proving (4.1). The zero-sum condition then kills the constant value on each nonzero residue class; the zero class was assumed zero. □

Lemma 4.2. *Assume the normalized setting of Proposition 3.7. Let $0 < a < n$ represent an element of* $\operatorname{supp} v$*, and let r, s, z be positive, off the d-grid, with $a + r + s + z = n$. Then*

$$\Delta K(a, r, s, z) = \frac{v(a)}{2} \sum_{\text{cyc}} \big(w(s) + w(z)\big)\big(\tau(r) - \tau(r+a)\big). \tag{4.2}$$

Its absolute value is at most two. If these differences vanish for every positive representative $a \in \operatorname{supp} v$ and every such triple r, s, z, then $\mathcal{B}_e$ is d-periodic.

Proof. The only proper subset sums on the d-grid are a and $n - a$. Thus a changed cut occurs only when the label a is first or last. For distinct first and last bottom labels x, y, pairing the two orders with a first or last gives the contribution

$$v(a)\big(b(a+x)b(-y) - b(x)b(-a-y)\big).$$

Off the d-grid, $b(-y) = b(y) - w(y)$ and $w(a+y) = w(y)$. Writing $\varphi_a(x) = (\tau(x) - \tau(x+a))/2 = b(x) - b(x+a)$ and summing over ordered first and last labels cancels the symmetric terms and gives (4.2).

At an active position $w(x) \neq 0$, both values of τ equal one. Otherwise both are even. Thus φ_a is ternary and $w\varphi_a = 0$. Among three disjoint ternary pairs $(\varphi_a(x), w(x))$, at most two cross-products are nonzero. This proves the integer bound, including repeated labels.

Fix $a \in \operatorname{supp} v$ and put $W = n - a$. On $\mathbb{Z}/W\mathbb{Z}$, define $\varphi_a(x) = (\tau(x) - \tau(x+a))/2$ for representatives $0 \leq x < W$ off the d-grid, and set $\varphi_a = 0$ on that grid. Since $e \mid d \mid W$, the function w descends to this circle with least period e: its word consists of repetitions of the same least period. Evenness of τ on the original circle gives $\tau(W-x) - \tau(W-x+a) = \tau(a+x) - \tau(x)$, so φ_a is odd on the smaller circle and $w\varphi_a = 0$.

Suppose every four-block difference in (4.2) vanishes. The balance equation (3.3) for w, φ_a on $\mathbb{Z}/W\mathbb{Z}$ follows by cases. If $x, y, x+y$ are off the d-grid, put $z = -x - y$. The positive representatives of x, y, z have total W or $2W$. In the first case the four-block identity gives

$$\sum_{\text{cyc}} \varphi_a(x)(w(y) + w(z)) = 0.$$

In the second, negate all three residues; both functions are odd, so the mixed products are unchanged. Substituting $z = -x - y$ gives (3.3) in both cases. If x is on the d-grid, then $w(x) = \varphi_a(x) = 0$ and $w(x+y) = w(y)$; all remaining terms vanish by $w\varphi_a = 0$. The case of y on the grid is the same. If $x + y$ is on the grid, then $w(y) = -w(x)$; either both w-values vanish or both corresponding φ_a-values vanish. Thus the equation holds everywhere. By Lemma 4.1, φ_a is constant on each nonzero residue class modulo e.

Choose between $a, n-a$ so that $0 < a < n/2$, possible because v is odd and vanishes at a self-opposite residue. Write $n = Me$, $a = te$, so $2t < M$. For $1 \leq r < e$, put $T_j = \tau(r + je)$, $0 \leq j < M$. The constancy just proved gives

$$T_j - T_{j+t} = 2c_r \qquad (0 \leq j < M - t)$$

for a constant c_r. If $w(r) \neq 0$, every $T_j = 1$. Otherwise every T_j belongs to $\{0, 2\}$, since these positions are off the d-grid and have zero odd part. The equations at $j = 0, t$ make T_0, T_t, T_{2t} an arithmetic progression in this two-element set. Hence $c_r = 0$ and all nonwrapped differences vanish.

Apply the same finite-orbit argument to the support point $n-a$, on its complementary circle of length $a = te$. It gives a constant c'_r with

$$T_j - T_{j+M-t} = 2c'_r \qquad (0 \le j < t).$$

The t wrapped differences for the original shift by t therefore all equal $-2c'_r$. Their sum is zero, because $\sum_{j=0}^{M-1}(T_j - T_{j+t \bmod M}) = 0$ and the nonwrapped terms already vanish. Thus $c'_r = 0$, and both shifts a and $-a$ preserve τ off the e-grid. Repeating this argument for each reflected pair in $\operatorname{supp} v$ covers every support shift. These shifts generate the d-grid; since $b = (\tau + w)/2$ there, $\mathcal{B}_e$ is d-periodic. □

4.2. **An algebraic chain cancellation.** For a finite label set M, give every proper nonempty vertex I of the Boolean lattice 2^M a weight $F(I)$ in a commutative ring. For $I \subseteq J$, let $Z_F(I, J)$ sum the products of intermediate vertex weights over maximal chains from I to J, omitting the endpoints. A chain with no intermediate vertex contributes one.

Lemma 4.3. *If F' differs from F only at vertices in an antichain $\mathscr{A}$, with increments $\delta_I = F'(I) - F(I)$, then*

$$Z_{F'}(\varnothing, M) - Z_F(\varnothing, M) = \sum_{I \in \mathscr{A}} \delta_I Z_F(\varnothing, I) Z_F(I, M).$$

Proof. A maximal chain meets at most one vertex of an antichain. Its weight therefore changes by the increment at that vertex times the unchanged weights elsewhere. Chains through I split uniquely into chains below and above it, giving the displayed summand. □

For a labeled coarsening, take $F(I) = b(\sum_{i \in I} a_i)$; the chain sum is (2.1). The two exceptional vertices below are $\{x\}$ and its complement, with opposite increments. They are incomparable. The surviving paths put x first or last, and deleting x leaves the same orders of the remaining labels, with or without translation by x. This gives the translated difference used in the next proof. Figure 1 illustrates the two path families.

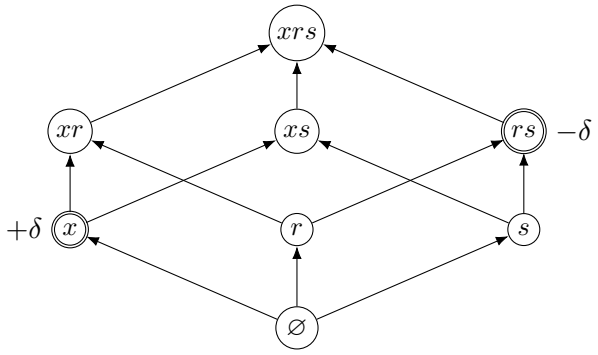


FIGURE 1. Complementary exceptional vertices. No maximal chain visits both. Deleting the distinguished label leaves the difference of two smaller path families. Numerical equality of labels does not identify vertices.

4.3. **The nested marked identity.** Fix a strict divisibility chain

$$e = p_0 < p_1 < \cdots < p_L = P \mid d, \qquad L \ge 1, \qquad \mathcal{B}_{p_i} \text{ is } p_{i+1}\text{-periodic} \quad (0 \le i < L). \tag{4.3}$$

These are period hypotheses, not least-period hypotheses. Take a positive labeled list

$$\mathbf{a} = (a, y_{L-1}, \ldots, y_0, r, s, z), \qquad \sum \mathbf{a} = n, \tag{4.4}$$

where $a \in \operatorname{supp} v$, the target is $t = y_{L-1}$, and

$$y_i \in p_i\mathbb{Z} \setminus p_{i+1}\mathbb{Z} \quad (0 \le i < L), \qquad e \nmid r, s, z. \tag{4.5}$$

When $L = 1$, every subsequence indexed from 0 to $L-2$ is empty and $y_0 = t$. Put

$$\begin{aligned} \Delta &= b(a+t) - b(t), \\ \theta_i &= \mathcal{B}_{p_{i+1}}(y_i + y_{i+1}) - \mathcal{B}_{p_{i+1}}(y_i) \quad (0 \le i \le L-2), \\ f_y(x) &= \mathcal{B}_e(x+y) - \mathcal{B}_e(x), \\ H_y(r,s,z) &= f_y(r)(w(s)+w(z)) + f_y(s)(w(r)+w(z)) + f_y(z)(w(r)+w(s)). \end{aligned} \tag{4.6}$$

For a labeled list $\mathbf{M} = (m_1, \dots, m_k)$, define

$$\mathcal{J}_y(\mathbf{M}) = \sum_{\pi \in \mathfrak{S}_k} \left[\prod_{j=1}^{k-1} b\left(y + \sum_{h=1}^{j} m_{\pi(h)} \right) - \prod_{j=1}^{k-1} b\left(\sum_{h=1}^{j} m_{\pi(h)} \right) \right]. \tag{4.7}$$

Theorem 4.4. *Under* (4.3)–(4.5),

$$\Delta K = -v(a)\Delta \left(\prod_{i=0}^{L-2} \theta_i \right) H_{y_0}(r,s,z). \tag{4.8}$$

The empty product is one. The absolute value is at most two, and the list has $L+4$ labels.

Proof. The sum $r+s+z$ is divisible by e. Every nonempty proper subset of these three labels has sum off the e-grid: for two labels, its sum is the negative of the remaining label modulo e. Adding higher labels does not change this. A subset containing no bottom label has a lowest neutral layer, whose contribution cannot be canceled by higher layers. If all bottom labels occur, apply the same observation to the complement. The only proper nonempty subset sums of the full list on the d-grid are therefore a and $n-a$.

Their increments from c to b are $v(a)$ and $-v(a)$. The antichain identity gives

$$\Delta K = v(a)\mathcal{J}_a(t, y_{L-2}, \dots, y_0, r, s, z).$$

In the remaining list, only $\{t\}$ and its complement have proper subset sums on the p_{L-1}-grid, by the same lowest-layer argument. At all other vertices the shift a preserves the cut value, since $P \mid d \mid a$ and $\mathcal{B}_{p_{L-1}}$ has period P. The two exceptional increments are Δ and

$$b(n-t) - b(n-a-t) = b(-t) - b(-a-t) = -\Delta.$$

Here $t, a+t$ lie on the e-grid and off the d-grid, where the cut values are even under reflection. Deleting the first or last label therefore leaves $v(a)\Delta\mathcal{J}_t(y_{L-2}, \dots, y_0, r, s, z)$.

For an intermediate step $1 \le i < L$, let

$$\mathbf{M}_i = (y_{i-1}, \dots, y_0, r, s, z), \qquad T_i = \sum \mathbf{M}_i, \qquad A_i = T_i + y_i = n - a - \sum_{k>i} y_k.$$

The integer A_i is divisible by p_{i+1}, and the total T_i is divisible by p_i. In $\mathbf{M}_i$, the only proper subset sums on the p_{i-1}-grid are those of $\{y_{i-1}\}$ and its complement. At every other vertex, shifting by y_i preserves the cut value because $p_i \mid y_i$ and $\mathcal{B}_{p_{i-1}}$ has period p_i. Write $x = y_{i-1}$. The singleton increment is θ_{i-1}. At the complement it is

$$\begin{aligned} b(T_i - x + y_i) - b(T_i - x) &= \mathcal{B}_{p_i}(A_i - x) - \mathcal{B}_{p_i}(A_i - y_i - x) \\ &= \mathcal{B}_{p_i}(-x) - \mathcal{B}_{p_i}(-y_i - x) \\ &= \mathcal{B}_{p_i}(x) - \mathcal{B}_{p_i}(x + y_i) = -\theta_{i-1}. \end{aligned}$$

Both arguments in the first line are off the p_i-grid. The second line uses period p_{i+1}, and the third uses the zero odd part at multiples of e. If $\mathbf{R}_i = (y_{i-2}, \dots, y_0, r, s, z)$, with $\mathbf{R}_1 = (r, s, z)$, antichain contraction gives

$$\mathcal{J}_{y_i}(y_{i-1}, \mathbf{R}_i) = \theta_{i-1}\mathcal{J}_{y_{i-1}}(\mathbf{R}_i).$$

This explicitly removes every intermediate label and leaves $\mathcal{J}_{y_0}(r,s,z)$ times the stated scalar factors.

At the bottom put $y = y_0$. Since $y + r + s + z \equiv 0 \pmod{p_1}$, the contribution with first label q_1 and last label q_2 is

$$\mathcal{B}_e(y+q_1)\mathcal{B}_e(-q_2) - \mathcal{B}_e(q_1)\mathcal{B}_e(-y-q_2).$$

Period p_1 justifies the complementary arguments. Using $\mathcal{B}_e(-x) = \mathcal{B}_e(x) - w(x)$ and $w(y+x) = w(x)$, this becomes

$$f_y(q_1)\mathcal{B}_e(q_2) - \mathcal{B}_e(q_1)f_y(q_2) - f_y(q_1)w(q_2).$$

The first two terms cancel on summing over the six ordered pairs of labels, even if numerical values coincide. The remaining sum is $-H_y(r,s,z)$, proving (4.8).

Finally, f_y, w are disjoint ternary functions: a nonzero value of w forces the binary cut value, which is unchanged by an e-multiple shift. On three positions at most two mixed products in H_y are nonzero. Thus $|H_y| \le 2$. Every other factor is zero or a unit, proving the bound. □

4.4. **Least periods select nonzero probes.**

Lemma 4.5. *Suppose $\mathcal{B}_e$ has least period $p > e$. If $e \mid y$ and $p \nmid y$, there are $0 < r, s, z < p$, all off the e-grid, such that*

$$r + s + z \equiv -y \pmod{p}, \qquad H_y(r,s,z) \neq 0. \tag{4.9}$$

Proof. The function $f_y = \mathcal{B}_e(\cdot + y) - \mathcal{B}_e$ on $\mathbb{Z}/p\mathbb{Z}$ is nonzero, vanishes on $e\mathbb{Z}$, has zero sum on every e-residue class, and satisfies $wf_y = 0$. Moreover,

$$f_y(-y - x) = -f_y(x). \tag{4.10}$$

Suppose $H_y(r,s,z) = 0$ for every triple satisfying $0 < r, s, z < p$, $e \nmid r, s, z$, and $r + s + z \equiv -y \pmod{p}$. Substituting $z = -y - r - s$, equation (4.10) and the oddness and e-periodicity of w give

$$w(r)f_y(s) + f_y(r)w(s) = w(r+s)(f_y(r) + f_y(s)) + f_y(r+s)(w(r) + w(s))$$

whenever $r, s, r+s$ are off the e-grid. If r is on that grid, then $w(r) = f_y(r) = 0$ and $w(r+s) = w(s)$, so the remaining products vanish by $wf_y = 0$. The case of s on the grid is identical. If $r + s$ is on the grid, then $w(s) = -w(r)$; either both w-values or both corresponding f_y-values vanish. Thus the equation holds on the whole circle. Lemma 4.1 and the zero residue-class sums would force $f_y = 0$, a contradiction. □

Lemma 4.6. *Suppose the periods in* (4.3) *are least periods, and fix $t \in p_{L-1}\mathbb{Z} \setminus P\mathbb{Z}$. The probes can be chosen so that*

$$0 < y_i < p_{i+2}, \quad \theta_i \neq 0 \quad (0 \le i \le L-2), \qquad 0 < r, s, z < p_1, \quad H_{y_0}(r,s,z) \neq 0, \tag{4.11}$$

with the layer conditions (4.5) *and*

$$t + \sum_{i=0}^{L-2} y_i + r + s + z \equiv 0 \pmod{p_1}. \tag{4.12}$$

Proof. Set $y_{L-1} = t$ and work downward. Once y_{i+1} is chosen, translation by it is not a period of $\mathcal{B}_{p_{i+1}}$, whose least period is p_{i+2}. The nonzero difference

$$\mathcal{B}_{p_{i+1}}(x + y_{i+1}) - \mathcal{B}_{p_{i+1}}(x)$$

vanishes on the p_{i+1}-grid. It also vanishes off the p_i-grid, where both mask values agree with $\mathcal{B}_{p_i}$ and the shift is a multiple of its period p_{i+1}. A nonzero value thus occurs in $p_i\mathbb{Z} \setminus p_{i+1}\mathbb{Z}$. The difference has period p_{i+2}; choose a representative $0 < y_i < p_{i+2}$ there. At the bottom $e \mid y_0$ but $p_1 \nmid y_0$, so Lemma 4.5 supplies the last three probes. When $L = 1$, it applies directly to $y_0 = t$. The bottom congruence cancels y_0, and every higher probe is divisible by p_1, proving (4.12). □

5. Realization at the original weight

The list selected by Lemma 4.6 initially has only the required congruence. It must be made into a positive list of total n before its chain sum is an inventory coefficient.

Lemma 5.1. *In the least-period setting of Lemma 4.6, let $0 < a < n$ represent an element of* $\operatorname{supp} v$*, and write*

$$W = n - a, \quad P = p_L, \quad \Sigma = \sum_{i=0}^{L-2} y_i + r + s + z, \quad C_{\mathrm{box}} = 3p_1 + \sum_{i=2}^{L} p_i. \tag{5.1}$$

Suppose $0 < t < W$ and $b(a+t) \neq b(t)$. If

$$W \geq C_{\mathrm{box}} - p_1, \tag{5.2}$$

then the selected probes or their reflected versions can be padded to a positive detector of total n, without changing the nonzero product in (4.8)*. In particular, $W \geq 2P$ suffices.*

Proof. Reflect the target and probes as in Fig. 2:

$$t' = W - t, \quad y'_{L-1} = t', \quad y'_i = p_{i+2} - y_i \quad (0 \leq i \leq L-2), \quad (r', s', z') = (p_1 - r, p_1 - s, p_1 - z).$$

All reflected probes remain positive and in the same divisibility layers. Their sum is $\Sigma' = C_{\mathrm{box}} - \Sigma$.

The original list fits if $\Sigma \leq W - t$. Otherwise the congruence (4.12) forces

$$\Sigma \geq W - t + p_1, \qquad \Sigma' \leq C_{\mathrm{box}} - W + t - p_1 \leq t.$$

So the reflected list fits. Strict divisibility gives $2p_1 + p_2 + \cdots + p_L \leq 2p_L$, which proves that $W \geq 2P$ suffices.

At each intermediate level $0 \leq i \leq L-2$, $y'_i \equiv -y_i$ and $y'_{i+1} \equiv -y_{i+1} \pmod{p_{i+2}}$. For $i = L-2$, the latter congruence uses $P \mid W$; at lower levels, it uses $p_{i+2} \mid p_{i+3}$. The mask $\mathcal{B}_{p_{i+1}}$ has that period and is even at these multiples of e, so

$$\theta'_i = \theta_i.$$

At the bottom, $f_{-y}(-x) = f_y(x)$ because the two odd parts cancel under an e-multiple shift, whereas $w(-x) = -w(x)$. Consequently $H_{y'_0}(r', s', z') = -H_{y_0}(r, s, z)$. Finally, $t, a+t$ are neutral off-d-grid positions, and

$$\Delta' := b(a+t') - b(t') = b(-t) - b(-a-t) = -\big(b(a+t) - b(t)\big).$$

The two minus signs cancel; the entire product retains its sign.

In the fitting orientation, the remaining deficit is a nonnegative multiple of p_1. Add it to one bottom probe. Every factor in the product is unchanged, and the final list satisfies

$$a + t + \sum_{i=0}^{L-2} y_i + r + s + z = n. \tag{5.3}$$

Use the primed variables in this equation when reflection was needed. The upper bounds in (4.11) apply to the initially selected probes; the identity itself requires only positivity, the layer conditions, and (5.3). □

Example 5.2. Let $n = 30$, $e = 5$, and $p_1 = d = 10$. Start with the periodic mask

$$(\mathcal{B}_5(0), \ldots, \mathcal{B}_5(9)) = (0, 1, 1, 0, 0, 0, 1, 0, 1, 0),$$

and set the remaining cuts by

$$b(5) = b(25) = 1, \quad b(15) = 0, \quad b(10) = 1, \quad b(20) = 0.$$

Let c agree with b except at $10, 20$, where the values are exchanged. Take $a = 10$, $W = 20$, and $t = 15$. The bottom probes $(9, 8, 8)$ give $H_t = 2$ and $\Delta = 1$, but their sum is 25, whereas $W - t = 5$. Reflection gives target 5, probes $(1, 2, 2)$, $H_5 = -2$, and $\Delta' = -1$. Padding the first probe by 10 gives the positive weight-30 list $(10, 5, 11, 2, 2)$. Its labeled difference is -2, and exact division by 2! gives ordinary coefficient difference -1 at $(11, 10, 5, 2, 2)$.

Proposition 5.3. *Suppose* (4.3) *consists of least periods. If all applicable positive detectors of Theorem 4.4 vanish, then $\mathcal{B}_P$ is d-periodic.*

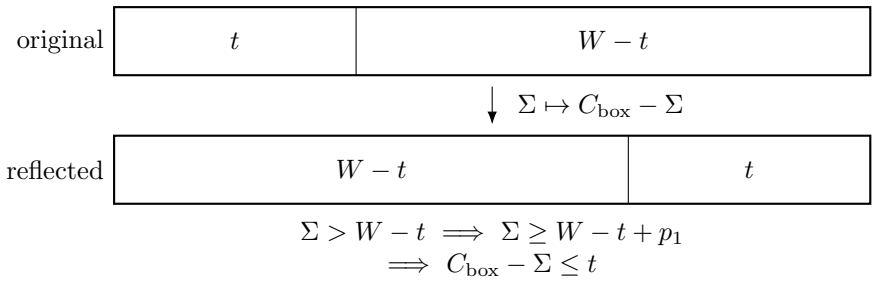


FIGURE 2. The reflection changes which complementary interval accommodates the probes. The lattice overshoot is at least p_1. The ambient weight remains $a + W = n$ throughout; the drawing is schematic, not to scale.

Proof. Outside the p_{L-1}-grid, invariance under any support shift $a \in \operatorname{supp} v$ follows from period $P \mid d$ of $\mathcal{B}_{p_{L-1}}$. Consider $t \in p_{L-1}\mathbb{Z} \setminus P\mathbb{Z}$ with $0 < t < n - a$. If $n - a \geq 2P$ and $b(a + t) \neq b(t)$, the nonzero probes of Lemma 4.6 and the positive realization of Lemma 5.1 give a nonzero detector, contrary to the hypothesis. Therefore

$$b(t + a) = b(t) \qquad (0 < t < n - a)$$

whenever the indicated layer and length conditions hold.

Suppose first that $P < d$, so $d \geq 2P$. For every support point a, both a and $n-a$ are positive multiples of d. Apply the nonwrapped equality to both support points. The first handles $0 < t < n - a$. If $n - a < t < n$, put $x = t - (n - a)$; then $0 < x < a$, and the complementary application gives

$$b(t) = b(x + n - a) = b(x) = b(t + a),$$

with the last argument read modulo n. The omitted endpoint $t = n - a$ lies on the P-grid and is not in the target layer. Thus every support shift preserves the cut values in that layer. The previous mask handles positions off the p_{L-1}-grid, and $\mathcal{B}_P$ is zero on the P-grid. Since the support generates $d\mathbb{Z}/n\mathbb{Z}$, the mask is d-periodic.

Now suppose $P = d$ and $d \in \operatorname{supp} v$. Choose $a = d$. The budget (5.2) holds because $n/d \geq 3$, hence $n - d \geq 2d$. For a residue $0 < r < d$ in $p_{L-1}\mathbb{Z} \setminus d\mathbb{Z}$, the nonwrapped equalities join every successive pair in

$$r,\ r + d,\ \ldots,\ r + (n/d - 1)d.$$

All their cut values agree, so the final cyclic wrap agrees too. Again the previous mask handles positions off the p_{L-1}-grid and $\mathcal{B}_d$ vanishes on its grid. This proves d-periodicity.

Finally suppose $P = d$ and $d \notin \operatorname{supp} v$. Oddness also excludes $n - d$. Every support representative and its complement are at least $2d$, so both directional budgets are available. The two-direction argument of the first case now applies with $P = d$. Every support shift preserves $\mathcal{B}_d$; since those shifts generate the d-grid, it is d-periodic. The three cases are exhaustive. □

Proposition 5.4. *Assume the starting mask and every subsequent propagation succeed. Set*

$$p_0 = e, \qquad p_{j+1} = \operatorname{per}(\mathcal{B}_{p_j}).$$

Then $p_j \mid p_{j+1} \mid d$ at every step. The chain stabilizes at a proper inner size p for which, after the chosen orientation,

$$\beta = \delta \circ \epsilon, \qquad \gamma = \zeta \circ \epsilon, \qquad 3 \leq |\epsilon| = p \leq d < n. \tag{5.4}$$

Proof. Every mask obtained in the iteration is d-periodic, so its least period divides d. Suppose the divisibilities have been proved through p_j, and let r be a period of $\mathcal{B}_{p_j}$. Its odd part is w, so $p_0 = e \mid r$. For $0 \leq k < j$, if $p_k \mid r$, setting the values on the p_k-grid to zero recovers $\mathcal{B}_{p_k}$, since $p_k \mid p_j$. Translation by r preserves that grid, so this remasking preserves period r. Consequently $p_{k+1} = \operatorname{per}(\mathcal{B}_{p_k}) \mid r$. Induction on k gives $p_j \mid r$; when $j = 0$, the odd-part argument already suffices. Apply this to $r = \operatorname{per}(\mathcal{B}_{p_j})$ to obtain $p_j \mid p_{j+1} \mid d$, completing the induction on j.

Each strict increase is followed by Proposition 5.3. A divisor chain inside d must stabilize. At its stable value p, $b(x + p) = b(x)$ for $p \nmid x$. The two originals agree outside the p-grid because

their changed support is contained in $d\mathbb{Z}$ and $p \mid d$. Let ϵ be the composition of weight p with cuts $S(\beta) \cap \{1, \dots, p-1\}$, and let δ, ζ have cuts $\{x : px \in S(\beta)\}$ and $\{x : px \in S(\gamma)\}$. Equation (2.4) proves the literal factorizations (5.4). The bounds follow from (3.7). □

Proposition 5.5. *For two compositions of weight n, either they are identical or reversed, or, after orientation, a proper literal common inner factor is found, or some ordinary coefficient differs by one or two. The initial tests use at most three blocks; failure of the starting mask uses four blocks; failure after $L \geq 1$ strict mask increases uses $L + 4$ blocks. In the last cases,*

$$L \leq \Omega(d/e), \qquad 4 + \Omega(d/e) \leq \Omega(n) + 2. \tag{5.5}$$

The distinguishing partition has distinct parts except possibly one pair.

Proof. Lemma 3.1 either supplies a small certificate or establishes exact low-order identities. Remove the identical and reversal cases, and normalize by Proposition 3.7.

If a special four-block expression in Lemma 4.2 is nonzero, it has labeled magnitude at most two and gives an ordinary certificate by Lemma 2.1. Otherwise the first mask is d-periodic. At a subsequent defect, Lemmas 4.6 and 5.1 and Theorem 4.4 supply an original-weight nonzero detector of labeled magnitude at most two. If no defect occurs, Proposition 5.4 supplies the factor.

A strict period increase consumes at least one prime factor of d/e. Since $e \geq 3$ and $n/d \geq 3$,

$$4 + \Omega(d/e) = \Omega(n) + 4 - \Omega(e) - \Omega(n/d) \leq \Omega(n) + 2.$$

The repeated-part assertion follows from Lemma 2.1. No full-inventory equality was used to obtain this alternative: only vanishing of the explicitly tested bounded discrepancies was used. □

6. Recognition by descent and composition congruence

Theorem 6.1 (Coarse recognition)**.** *For compositions β, γ of weight n, either they have a factor-reversal witness and $\Psi_\beta = \Psi_\gamma$, or some $\lambda \vdash n$ satisfies*

$$\ell(\lambda) \leq \Omega(n) + 2, \qquad m_\beta(\lambda) - m_\gamma(\lambda) \in \{-2, -1, 1, 2\}. \tag{6.1}$$

The alternatives are exclusive. The certificate may have distinct parts except possibly one repeated pair.

Proof. Use strong induction on n, with both the witness and equality of the whole substitution maps as the successful output. Literal equality and reversal give this output directly by Proposition 2.4. In every other case, Proposition 5.5 either supplies (6.1) or gives

$$\beta = \delta \circ \epsilon, \qquad \eta = \zeta \circ \epsilon, \qquad \eta \in \{\gamma, \gamma^*\}, \qquad p = |\epsilon| > 1.$$

Apply the induction hypothesis to the outer pair of weight n/p. For any partition $\mu \vdash n/p$, retaining only grid cuts gives

$$m_\beta(p\mu) = m_\delta(\mu), \qquad m_\eta(p\mu) = m_\zeta(\mu). \tag{6.2}$$

Thus an outer certificate lifts unchanged in magnitude and number of blocks, and $\Omega(n/p) + 2 \leq \Omega(n) + 2$.

Otherwise induction gives $\Psi_\delta = \Psi_\zeta$ and a factor witness. Compose the maps on the left with Ψ_ϵ and use (2.7). Append ϵ to the witness. If $\eta = \gamma^*$, reverse every factor on the second side, using $(\alpha \circ \epsilon)^* = \alpha^* \circ \epsilon^*$; the order of the factor list is not reversed. This gives the desired witness and map equality for the original pair.

A factor witness preserves the complete inventory by Corollary 2.5, so the alternatives are exclusive. Scaling in (6.2) preserves repeated-part multiplicities. □

Remark 6.2. The outer pair is not assumed fully equivalent at extraction. It receives the induction alternative. In a formulation starting from shallow congruence, (6.2) transfers that shallow congruence, and the smaller-weight theorem then gives full equality.

Proposition 6.3. *Let $H : \mathbb{N} \to \mathbb{N}$ be monotone under divisibility. Suppose it covers the initial low-order tests and every applicable local detector in Proposition 5.5: three blocks for $n \geq 5$, two blocks for $2 \leq n \leq 4$, four blocks whenever the starting mask requires a test, and $L + 4$ after $L \geq 1$ strict mask increases requiring a detector. Then Theorem 6.1 holds with $H(n)$ in place of $\Omega(n) + 2$.*

Proof. Repeat the same induction, using $H(n/p) \leq H(n)$ in (6.2). At weights at most four, two-block data fix every symmetric cut pair and the midpoint, with at most one asymmetric reflected pair left; its two orientations are global reversals. Weight one has a single composition. The other local costs are exactly the stated detector costs. □

Proof of Theorem 1.2. If inventories are equal, the certificate alternative in Theorem 6.1 is impossible. Conversely, equality of the maps gives equality on h_1, and a factor witness preserves G. If $\alpha \sim \alpha'$ and $\beta \sim \beta'$, then

$$\alpha \circ \beta \sim \alpha' \circ \beta'. \tag{6.3}$$

For (6.3), use $\Psi_{\alpha\circ\beta} = \Psi_\beta \Psi_\alpha$. Equality on all generators gives every near-power simultaneously. □

Thus composition descends to the equivalence classes. The class of ϵ determines Ψ_ϵ, and equality of these maps reflects equality of classes by evaluation at h_1. The composition law reverses the order of the corresponding endomorphisms. For a fixed right factor, the grid left inverse also reflects equivalence of the outer factors; no choice of a representative of an inner class is required.

Corollary 6.4. *For every $m \geq 1$,*

$$\mathcal{E}((m) \circ \beta) = \{(m) \circ \gamma : \gamma \in \mathcal{E}(\beta)\}.$$

Proof. The inclusion from right to left is (1.4). For the reverse inclusion, use strong induction on $N = mn$, where $n = |\beta|$. Put $F = \beta^{\odot m}$, and suppose $\eta \sim F$. If $\eta = F$ or F^*, take $\gamma = \beta$ or β^*. Otherwise Proposition 5.5, with its certificate excluded by inventory equality, gives

$$F = \alpha \circ \epsilon, \qquad \eta' = \zeta \circ \epsilon, \qquad \eta' \in \{\eta, \eta^*\}, \qquad 1 < p = |\epsilon| < N.$$

By (2.9), $G_\alpha = G_\zeta$.

Suppose first that $p \nmid n$. Let B be the cyclic cut indicator of F on $\mathbb{Z}/N\mathbb{Z}$. It is n-periodic, and its values off the p-grid depend only on the residue modulo p. If $j \in S(\alpha)$, then $B(pj) = B(pj+n) = 1$. Since $pj+n$ and n are off the p-grid and congruent modulo p, this gives $B(n) = 1$, contrary to the absence of near-power seam cuts. Thus $\alpha = (N/p)$. Its inventory has ordinary degree one, so $G_\zeta = G_\alpha$ forces $\zeta = \alpha$. Hence $\eta' = F$, and the conclusion follows.

Now suppose $p \mid n$, and put $k = n/p$. Restricting the cut formula to the first n-cell and applying Lemma 2.3 gives $\beta = \delta \circ \epsilon$. Equation (2.4) and uniqueness at inner weight p then give $\alpha = \delta^{\odot m}$. Since $mk = N/p < N$, the induction hypothesis applied to $\zeta \sim \delta^{\odot m}$ gives

$$\zeta = \xi^{\odot m}, \qquad \xi \sim \delta.$$

Set $\gamma = \xi \circ \epsilon$. By (2.6), $\gamma \sim \beta$, and the cut formula gives $\eta' = \gamma^{\odot m}$. If $\eta' = \eta^*$, replace γ by γ^*. Literal reversal preserves G and commutes with near-powers. □

Corollary 6.5. *Under a budget H in Proposition 6.3, ordinary inventory coefficients through $H(n)$ modulo any integer $q \geq 3$ determine the complete inventory.*

Proof. A nonzero discrepancy of magnitude one or two is nonzero modulo q. Apply the exact integer alternative. No labeled factorial is inverted modulo q. □

7. Arithmetic bounds and sharp cumulative depth

7.1. **Restrictions on period growth.**

Lemma 7.1. *In the normalized mask argument:*

(1) *if $e = 3$, a common inner factor of weight three is immediate;*

(2) *if $e = 4$, the first strict mask period cannot be eight;*
(3) *a strict chain of successive least mask periods cannot contain $p, 2p, 4p$ consecutively.*

Proof. For $e = 3$, the nonzero odd ternary function w is nonzero on both nonzero residue classes. The corresponding binary cut values are forced: $b = 1$ where $w = 1$ and $b = 0$ where $w = -1$. They agree for the originals and repeat modulo three. The cuts on the three-grid are arbitrary outer cuts, giving the common inner factor without a four-block test.

For $e = 4$, a nonzero odd function of period four is nonzero on the odd residues and zero on residue two. If $\mathcal{B}_4$ had least period eight, its values at $1, 5$ and at $3, 7$ would agree by the forced odd bits. Its values at $2, 6$ agree by evenness at those neutral positions; it is zero at $0, 4$. It would therefore already be four-periodic.

Suppose $\mathcal{B}_p$ has period $2p$ and $\mathcal{B}_{2p}$ has least period $4p$. Off the p-grid, $\mathcal{B}_{2p} = \mathcal{B}_p$, so it is $2p$-periodic there. On the p-grid modulo $4p$, the values at $0, 2p$ are zero, and those at $p, 3p$ agree by (3.9), since $e \mid p$. Thus $\mathcal{B}_{2p}$ is $2p$-periodic everywhere, a contradiction. □

For integers $e \geq 4$ and $D \geq 1$, let $a_2 = \nu_2(D)$ be the exponent of 2 in D, and put $a_o = \Omega(D) - a_2$. Define

$$\varepsilon_e = \begin{cases} 0, & e = 4, \\ 1, & e \geq 5, \end{cases} \qquad T_e(D) = \min\left\{ a_2 + a_o, \left\lfloor \frac{2a_2 + 4a_o + \varepsilon_e}{3} \right\rfloor \right\}. \tag{7.1}$$

Lemma 7.2. *Let $e \geq 4$, $D \geq 1$, $a_2 = \nu_2(D)$ and $a_o = \Omega(D) - a_2$. The largest length of a list of integers at least two whose product divides D, with no adjacent entries equal to two and with the first entry different from two when $e = 4$, is $T_e(D)$ in* (7.1).

Proof. Write $b = a_o$ and $\varepsilon = \varepsilon_e$. If k entries equal two and q_0 entries are different from two, then

$$k \leq q_0 + \varepsilon.$$

If t_0 of the latter entries have no odd prime factor, they use at least two factors of two each. Thus

$$a_2 \geq k + 2t_0, \qquad b \geq q_0 - t_0,$$

and

$$2a_2 + 4b \geq 2k + 4q_0 \geq 3(k + q_0) - \varepsilon.$$

Also $k + q_0 \leq a_2 + b$. These prove the two upper bounds in (7.1).

For attainability, treat the odd primes with multiplicity as separate non-two entries. If $a_2 \leq b + \varepsilon$, all primes may remain separate and the singleton twos can be interspersed in the permitted slots; this attains $a_2 + b$.

Suppose $a_2 > b + \varepsilon$ and write $a_2 - b - \varepsilon = 3t_0 + r$, $r \in \{0, 1, 2\}$. If $r = 0$ or 1, use t_0 entries equal to four, the b odd primes, and $b + t_0 + \varepsilon$ singleton twos. At most one factor of two is unused. The resulting length is $2b + 2t_0 + \varepsilon$, as required. If $r = 2$, use $t_0 + 1$ fours, the b odd primes, and $b + t_0 + \varepsilon$ singleton twos. This has the required length one larger. In either case the twos can be placed in nonadjacent slots, with no initial two when $\varepsilon = 0$. The product is allowed to divide D, so unused prime factors cause no problem. The empty list covers the zero-length cases. □

Lemma 7.2 proves that $T_e(D)$ is the largest length of a numerical list of ratios whose product divides D, with no consecutive ratios equal to two and with the first ratio different from two when $e = 4$. This numerical optimization is separate from any assertion about attaining a recognition bound at a particular weight.

Definition 7.3. Set $B(1) = 1$ and $B(2) = B(3) = B(4) = 2$. For $n \geq 5$, define

$$B(n) = \min\left\{ n,\ \max\left(\{3\} \cup \{ 4 + T_e(d/e) : e \mid d \mid n,\ e \geq 4,\ n/d \geq 3 \} \right) \right\}. \tag{7.2}$$

An empty set of divisor candidates contributes nothing to the maximum.

Proposition 7.4. *The function B is monotone under divisibility and covers every applicable detector. For every n, $B(n) \leq \Omega(n) + 2$. Consequently the discrepancy and modular conclusions of Theorem 1.1 hold.*

Proof. An actual chain with L strict steps has ratio product dividing d/e and obeys Lemma 7.1. Thus $L \le T_e(d/e)$, and the required $L+4$ blocks are covered by (7.2). The starting four-block test is the case $L = 0$. The $e = 3$ case is immediate; the low-order tests and the elementary weights at most four are explicitly covered. No coarsening uses more than n blocks, so the outer minimum by n is harmless.

If $m \mid n$, every divisor candidate $e \mid d \mid m$ is also a candidate for n. Both entries in the outer minimum are nondecreasing under this passage, and the small-weight normalizations preserve monotonicity. Finally, $T_e(d/e) \le \Omega(d/e)$, so (5.5) gives the coarse upper bound. Apply Proposition 6.3 and Corollary 6.5. □

Some resulting formulas, with the normalization in Definition 7.3, are

$$\begin{aligned} B(2^a) &= 4 + \left\lfloor \frac{2(a-4)}{3} \right\rfloor && (a \ge 4), \\ B(3^a) &= a + 1 && (a \ge 2), \\ B(p^a) &= a + 2 && (p \ge 5 \text{ prime},\ a \ge 1). \end{aligned}$$

For powers of two, e and n/d each consume at least two prime factors; the remaining ratio count obeys the nonadjacent-two restriction. For powers of three, e consumes at least two factors and n/d at least one. For $p \ge 5$, each consumes one. These formulas concern sufficient arithmetic cutoffs, not proved exact values of $D_=$ at every prime power.

7.2. **The weight-only upper bound.** Let $U(n)$ denote the right-hand side of (1.6), without assuming the asserted equality with $C(n)$.

Proposition 7.5. *For $n \ge 2$, one has $B(n) \le U(n)$. For every $n \ge 1$,*

$$B(n) \le 2 \log_6 n + \log_6 9.$$

The additive constant cannot be decreased uniformly over all positive weights.

Proof. Fix $n \ge 5$ and a divisor candidate $e \mid d \mid n$ in (7.2), and put $L = T_e(d/e)$. By Lemma 7.2, there is a list $r_1, \dots, r_L \ge 2$ whose product divides d/e, with no consecutive twos and with the first entry different from two when $e = 4$. Thus

$$n = \frac{n}{d}\, e\, \frac{d}{e} \ge 3e \prod_{i=1}^{L} r_i.$$

If $L = 2j$, at least j ratios are at least three, so their product is at least 6^j. Since $e \ge 4$,

$$n \ge 12 \cdot 6^j. \tag{7.3}$$

If $L = 2j+1$ and $e = 4$, the first ratio is not two, so the product is at least $3 \cdot 6^j$, giving $n \ge 36 \cdot 6^j$. For $e \ge 5$, its minimum is $2 \cdot 6^j$, giving the sharper possible threshold

$$n \ge 30 \cdot 6^j. \tag{7.4}$$

These inequalities include $L = 0$, where the first threshold is twelve. They imply $4 + T_e(d/e) \le U(n)$ for every divisor candidate. The default value three is also at most $U(n)$ for $n \ge 5$, and the outer minimum in (7.2) can only lower the result. Hence $B(n) \le U(n)$. For $2 \le n \le 4$, both values equal two.

For the logarithmic estimate, the same thresholds give

$$\frac{6^{L+4}}{n^2} \le \begin{cases} 9, & L = 2j, \\ 216/25 < 9, & L = 2j+1. \end{cases}$$

The default value three satisfies $6^3/n^2 \le 216/25 < 9$ for $n \ge 5$. Taking the maximum over the candidates, then the outer minimum, proves $6^{B(n)} \le 9n^2$. The prescribed values at $n \le 4$ satisfy the same inequality. Taking logarithms gives the asserted bound, with equality at $n = 2$, proving optimality of the uniform additive constant. □

By Propositions 7.4 and 7.5, every unequal pair of weight $n \geq 2$ has a distinguishing coefficient through degree $B(n) \leq U(n)$. Since U is nondecreasing, this gives $C(N) \leq U(N)$; at weight one the maximum is empty. Thus U is a sufficient weight-only cutoff. Its noninitial transitions are $12, 30, 72, 180, 432, 1080, \ldots$, and its explicit formula gives $U(n) = 2\log_6 n + O(1)$. Sharpness requires actual unequal compositions, not just numerical ratio lists.

7.3. **A lift with an all-odd induction invariant.** For compositions R, S of common weight w, let

$$\mathrm{Alt}_j(R,S) = R \odot S \odot R \odot \cdots \odot S \odot R$$

have $2j+1$ factors. Define

$$\mathsf{OA}_s(R,S): \quad \text{for every } j \geq 0, \quad \left(G_{\mathrm{Alt}_j(R,S)} - G_{\mathrm{Alt}_j(S,R)}\right)^{[<s]} = 0. \tag{7.5}$$

The quantifier over every odd length is essential. The assertion for $j = 0$ alone would not control the odd segments produced in the next lift.

Set

$$E = R \odot S, \qquad F = R \cdot S, \qquad R' = E \cdot (F \odot E), \qquad S' = (E \odot F) \cdot E. \tag{7.6}$$

Both new roots have weight $6w$.

Theorem 7.6. *For $s \geq 2$,*

$$\mathsf{OA}_s(R,S) \Longrightarrow \mathsf{OA}_{s+2}(R',S'). \tag{7.7}$$

Moreover,

$$\partial_{h_{2w}}\partial_{h_{3w}}(G_{R'} - G_{S'}) = G_R - G_S. \tag{7.8}$$

If moreover $\mathsf{OA}_s(R,S)$ holds and $G_R - G_S$ first differs in degree s, with an ordinary unit witness at $\lambda \vdash w$, then $G_{R'} - G_{S'}$ first differs in degree $s+2$ and has the same unit witness at $\mathrm{sort}(\lambda \cup \{2w, 3w\})$.

Proof. Each lifted factor consists of the six old cells R, S, R, S, R, S. Its available cell boundaries are $\{2,3\}$ for R' and $\{3,4\}$ for S'. In a near-concatenation of an odd number of alternating lifted factors, boundaries between lifted factors are absent. Reflection through the midpoint exchanges the two available boundary patterns: reflection preserves the parity of the lifted-factor index and exchanges local boundaries two and four.

Group coarsenings by the old cell-boundary cuts retained. Once these cuts are chosen, every segment is an alternating near-concatenation of old cells, and its internal coarsening inventory is an independent factor. If only common odd boundaries are retained, the same retained subset occurs on both sides and gives literally the same product. If only even boundaries are retained, every segment has an even number of cells. Reflection reverses the order of the segments and preserves the alternating word in each such segment.

If both parities occur, at least two cuts are retained, giving at least three nonempty segments. Reflection preserves each even-cell segment and exchanges the two orientations of each odd-cell segment. The hypothesis (7.5) applies to every odd segment, whatever its length. In the telescoping identity for the corresponding segment inventories P_i, Q_i,

$$\prod_{i=1}^{t} P_i - \prod_{i=1}^{t} Q_i = \sum_{j=1}^{t} (P_j - Q_j) \prod_{i<j} P_i \prod_{i>j} Q_i,$$

each nonzero summand has one difference starting in degree at least s and $t-1$ nonempty factors starting in degree at least one. Since $t \geq 3$, it starts in degree at least $s+2$. This proves (7.7) for every odd number of lifted factors. No composition compatibility theorem is used.

Expanding the two alternative seams gives

$$G_{R'} - G_{S'} = G_E \left(G_R G_{S\odot R\odot S} - G_{R\odot S\odot R} G_S\right). \tag{7.9}$$

A composition of weight t has the term h_t once and no other term containing h_t. Since $|E| = 2w$, the h_{3w} derivative of (7.9) is $G_E(G_R - G_S)$. The h_{2w} derivative is therefore (7.8).

Every part of λ is at most w. The added parts $2w, 3w$ are distinct, exceed all old parts, and each occurs once. Thus the derivative transports the ordinary coefficient unchanged, with no factorial conversion. The low-degree vanishing and the surviving unit witness prove the final assertion. □

Proof of Theorem 1.3. For the even family, start with $R_0 = (1, 1)$ and $S_0 = (2)$. Their difference is h_1^2. Every nonempty composition of a specified weight has the same one-block term, so $\mathsf{OA}_2(R_0, S_0)$ holds at every odd alternating length. At each iteration, (7.7) supplies the all-odd hypothesis for the next pair before the exact-degree clause of Theorem 7.6 is applied. After r lifts, the weight is $2 \cdot 6^r$ and the first discrepancy is $2r + 2$, with a unit witness. For $r = j + 1$, these attain (7.3).

For the odd family, start with $R_0 = (1, 1, 3)$ and $S_0 = (1, 2, 2)$. Their difference is $h_1^2 h_3 - h_1 h_2^2$. They have identical two-block data. In a near-concatenation of an odd number of equal-weight alternating cells, reflection preserves the parity of the cell index. A two-block count therefore reduces to the sum of the reflected cut bits within a single root, which is equal for these seeds; seams are absent on both sides. Thus $\mathsf{OA}_3(R_0, S_0)$ holds for every odd length. Iterating (7.7) supplies OA_{3+2r} for each pair after r lifts. The exact-degree clause of Theorem 7.6 therefore gives weight $5 \cdot 6^r$ and first discrepancy $2r + 3$, again with a unit witness. For $r = j + 1$, these attain (7.4).

The initial witnesses are $(1, 1)$ and $(3, 1, 1)$, respectively. At every lift append the parts $2w, 3w$. These give pairs at each transition with no lower-degree discrepancy and a discrepancy of $+1$ at the transition. Combining with the upper bounds proves (1.6), including $N = 1$ and the elementary initial intervals. Unit witnesses remain nonzero modulo every $q \geq 3$, proving the modular sharpness statement as well. The comparison in Proposition 7.5 now also gives $B(n) \leq C(n)$ for $n \geq 2$. □

Remark 7.7. The exact theorem concerns $C(N)$, not $D_=(n)$ at every weight. Equality $D_=(n) = B(n)$ is not asserted. The depth theorem is stated for unsigned inventories. In a direct statement about signed ribbon h-coefficients, the coefficient of h_n first records the parity of the composition length. For example, the unsigned pair $(1, 1), (2)$ first differs in degree two, whereas their signed ribbon difference already has an h_2 term. The lifted threshold pairs have equal lengths, so that small endpoint issue does not affect the threshold families.

8. Composition coordinates and effective certificates

8.1. **Integral coordinates.**

Proposition 8.1. *Fix ϵ of weight p. Define*

$$\Theta_\epsilon(h_r) = \begin{cases} G_{\epsilon^{\odot(r/p)}}, & p \mid r, \\ h_r, & p \nmid r. \end{cases}$$

Then Θ_ϵ is a weight-preserving automorphism of $\Lambda_\mathbb{Z}$. In particular,

$$\Lambda_\mathbb{Z} = \mathbb{Z}\big[G_\epsilon, G_{\epsilon^{\odot 2}}, \ldots, h_r \ (p \nmid r)\big] \tag{8.1}$$

is a polynomial ring in the displayed algebraically independent generators. The map $\Psi_\epsilon \pi_p$ is an idempotent retraction onto $\operatorname{im} \Psi_\epsilon$.

Proof. For each j,

$$G_{\epsilon^{\odot j}} = h_{pj} + R_j(h_1, \ldots, h_{pj-1}).$$

Every term other than the one-block term has all indices smaller than pj. Thus Θ_ϵ is unitriangular in the ordered variables, with an integral inverse constructed recursively. This proves (8.1). The retraction assertion follows from $\pi_p \Psi_\epsilon = \mathrm{id}$. □

Proposition 8.2. *For a nonzero polynomial F whose first ordinary degree is r,*

$$(\Psi_\epsilon F)^{[r]} = F^{[r]}(h_j \mapsto h_{pj}). \tag{8.2}$$

Thus fixed-inner composition preserves first distinguishing degree over any commutative coefficient ring. Over $\mathbb{Z}$, it also preserves the gcd of all coefficients.

Proof. Each generator image is h_{pj} plus terms of ordinary degree at least two, proving (8.2). Relabeling the independent variables is injective over any coefficient ring. Integrality gives $\mathrm{cont}(F) \mid \mathrm{cont}(\Psi_\epsilon F)$, while the integral left inverse gives the reverse divisibility. □

8.2. **Recovery at prescribed factor weights.** For a composition F, let P_F be the sum of G over all proper prefixes, including the empty prefix, and let Q_F be the analogous suffix sum. For $1 \le j \le |F|$, distinguishing a block of size j gives

$$\partial_{h_j} G_F = [P_F Q_F]_{\mathrm{wt}=|F|-j}.$$

The two exterior pieces are disjoint at this weight, so they can be coarsened independently.

Theorem 8.3. *Let $|\alpha| = m$ and $|\beta| = n$. Then*

$$\pi_n(G_{\alpha\circ\beta}) = G_\alpha, \tag{8.3}$$

$$\partial_{h_{(m-1)n+j}} G_{\alpha\circ\beta} = \partial_{h_j} G_\beta \qquad (1 \le j \le n). \tag{8.4}$$

Consequently the rational-linear operator

$$\mathcal{L}_{m,n} = \frac{1}{n} \sum_{j=1}^{n} j h_j \partial_{h_{(m-1)n+j}} \tag{8.5}$$

satisfies $\mathcal{L}_{m,n}(G_{\alpha\circ\beta}) = G_\beta$. For displayed products with the same two factor weights,

$$\alpha \circ \beta \sim \alpha' \circ \beta' \quad \Longleftrightarrow \quad \alpha \sim \alpha' \text{ and } \beta \sim \beta'. \tag{8.6}$$

Proof. The outer recovery is the split substitution identity. For the derivative identity, first suppose $m \ge 2$. Differentiation marks one coarsened block of length $(m-1)n + j$ and removes its variable. Its exterior weight is $n - j < n$. If its left endpoint is x, then $0 \le x \le n - j$, and its right endpoint is $x + (m-1)n + j$, in the final n-cell or at the total endpoint. The literal cut formula identifies the allowed cuts before it with the prefix cuts of β up to x, and the allowed cuts after it with the suffix cuts after $x + j$, translated by $(m-1)n$. Deleting the central length $(m-1)n$ therefore gives a marked j-block in a coarsening of β, preserving every exterior block. Conversely, insert that length into any marked j-block. These operations are inverse and preserve the monomial in the unmarked blocks. Repeated blocks are counted with their marked multiplicity, exactly as in differentiation. This proves (8.4). For $m = 1$, the outer composition is (1), and the identity is literal.

Weighted Euler gives $\sum_{j=1}^{n} j h_j \partial_{h_j} G_\beta = n G_\beta$. Substitute (8.4) and divide by n over $\mathbb{Q}$ to obtain (8.5). The two recovery operators prove the forward implication of (8.6); its reverse is (6.3). □

These operators recover inventories of the factors at prescribed weights. They neither discover a factorization nor recover unique literal factor words. The division by n in (8.5) is over $\mathbb{Q}$; no such division is asserted over an arbitrary coefficient ring. The integral derivative identities themselves need no division.

8.3. **Checking a short certificate.** Let $t_1, \dots, t_s$ be the distinct part values of a partition $\lambda \vdash n$, with multiplicities $\nu_1, \dots, \nu_s$, and let $r = \ell(\lambda)$. For a state $a = (a_1, \dots, a_s)$, $0 \le a_i \le \nu_i$, put $s(a) = \sum_i a_i t_i$. Write $\nu = (\nu_1, \dots, \nu_s)$, and let e_i be the ith coordinate vector. Define $D(0) = 1$ and, for $a \ne 0$,

$$D(a) = \eta_b(a) \sum_{i : a_i > 0} D(a - e_i), \qquad \eta_b(a) = \begin{cases} 1, & a = \nu, \\ b(s(a)), & a \ne \nu. \end{cases}$$

Then

$$m_\beta(\lambda) = D(\nu). \tag{8.7}$$

Indeed, partition the distinct orders of the selected multiset according to their last part value. Each predecessor contributes once, since equal occurrences are not separately labeled. The cut test enforces the final partial sum except at the full state, where that sum is the endpoint n. Induction over the number of selected parts proves the recurrence.

There are $\prod_i(\nu_i+1) \le 2^r$ states and at most r terms per state, so the count takes $O(r2^r)$ arithmetic operations and $O(2^r)$ integer storage after the cut array is available. Since the recurrence uses only addition and binary cut tests, it also computes the ordinary coefficient directly modulo any integer, without division. The labeled alternative counts K_b on subsets of r labels; that version requires exact division by $\kappa(\lambda)$ before modular reduction. A certificate of Theorem 1.1 is thus checked on its short block list rather than by expanding all coarsenings of the original composition.

8.4. **A cut-array recognition algorithm.** The following deterministic procedure avoids expanding inventories. At each stage the two compositions have the same weight.

(1) If their cut words are equal or reflected, accept and record that literal operation.
(2) For both orientations of the second word, test divisors p of the current weight for a literal common inner factor: both words must be p-periodic off $p\mathbb{Z}$ and agree there.
(3) If neither orientation has a common factor of weight $p > 1$, reject. Otherwise choose the largest available p, remove that common factor, and continue with the outer pair on the p-grid.

Every removal is safe by the split embedding (2.9). Completeness follows from the factor alternative: an equivalent pair which is neither identical nor reversed has a proper common inner factor in one orientation. Reversals and removed factors reconstruct the witness on acceptance.

On an expanded n-entry cut array, testing one divisor takes $O(n)$ operations. At recursive weights n_i, one has $n_{i+1} \le n_i/2$ and $\sigma_0(n_i) \le \sigma_0(n)$, where $\sigma_0(n)$ is the number of positive divisors of n. Hence total time is

$$O\left(\sum_i n_i\sigma_0(n_i)\right) = O(n\sigma_0(n)).$$

This is an arithmetic-operation bound in the expanded cut-array model. It is not a polynomial bound in the binary encoding length of large composition parts. Certificate checking via (8.7) uses ordinary counts, either integrally or directly modulo three.

8.5. **The characteristic-two boundary.**

Proposition 8.4. *Let*

$$U_{17} = (1,1,1,1,1,1,1,1,9), \qquad V_{17} = (1,1,2,4,2,1,1,2,3).$$

Then

$$G_{U_{17}}^{[\le 3]} \equiv G_{V_{17}}^{[\le 3]} \pmod 2, \qquad G_{U_{17}} \ne G_{V_{17}}.$$

Specifically, the coefficients at $(14,2,1)$ *are two and zero, while those at* $(14,1,1,1)$ *are one and zero.*

Proof. Both cut sets contain exactly one point of every reflected pair modulo 17. They are

$$\{1,2,3,4,5,6,7,8\}, \qquad \{1,2,4,8,10,11,12,14\}.$$

Their disagreement set is $\{3,5,6,7,10,11,12,14\}$, a complete doubling-negation orbit modulo 17.

For any odd weight and cut indicator with $b(x)+b(-x)=1$ off zero, a three-part partition with distinct parts and k cut bits equal to one has ordinary multiplicity $k(3-k) \in \{0,2\}$. Indeed, an order contributes $b(\text{first})(1-b(\text{last}))$. For a repeated partition $(a,a,n-2a)$, its coefficient is one precisely when $b(a) = b(2a)$, and zero otherwise. This follows by writing its three possible numerical orders; the middle order contributes $b(a)b(n-a) = 0$. If all three parts coincide, both descriptions give zero. Thus two such indicators have equal three-block data modulo two exactly when their disagreement bits are constant on doubling orbits. Their one- and two-block data already agree. This proves the congruence.

For U_{17}, the ordered coarsenings realizing $(14,2,1)$ are $(1,2,14)$ and $(2,1,14)$; none occur for V_{17}. The partition $(14,1,1,1)$ occurs only as $(1,1,1,14)$ for U_{17} and not for V_{17}. □

Here $B(17) = 3$. Thus modulo two cannot replace modulo three at the stated arithmetic cutoff, nor can the discrepancy bound be replaced uniformly by a unit discrepancy at that cutoff. This example is nevertheless distinguished modulo two at degree four. It says nothing by itself about equality of *full* inventories modulo two.

Corollary 8.5. *For an odd prime q, two individual ribbon Schur functions become equal over $\mathbb{F}_q$ if and only if they were equal over $\mathbb{Z}$.*

Proof. The coefficient of h_n in (2.3) first forces the same length parity, because $1 \neq -1$ in odd characteristic. Ribbon equality then gives $G_\beta \equiv G_\gamma \pmod q$. Apply Theorem 1.1. This is not an assertion about arbitrary linear combinations of ribbons. □

9. Concluding remarks

The recursive lower families show that the logarithmic depth cannot be replaced by a bounded uniform depth.

For general skew diagrams, composition operations alone do not supply the reconstruction argument. Analogues of support separation, literal factor extraction, and positive detection inside the original diagram would be needed. The present results establish these ingredients for ribbons only.

Question 9.1. What is the exact maximum $D_=(n)$ of the first distinguishing depth at each fixed weight n, and for which weights is $B(n)$ attained? The cumulative frontier determines this at its transition weights, but not at every individual weight.

Question 9.2. Does equality of full coarsening inventories modulo two imply equality over the integers? If so, what block depth suffices for parity recognition? The weight-seventeen example is distinguished in degree four and therefore does not settle this question.

Appendix A. A multilevel marked detector

The following example uses two strict mask-period increases. It combines the antichain cancellation of Theorem 4.4 with the reflected packing of Lemma 5.1; neither statement depends on the example.

Let $n = 72$ and define

$$\begin{aligned} S = {} & \{x : 1 \le x < 72,\ x \equiv 1 \pmod 4\} \\ & \cup \{x : 1 \le x < 72,\ x \equiv 6 \pmod{12}\} \\ & \cup \{x : 1 \le x < 72,\ x \equiv 8 \text{ or } 16 \pmod{24}\} \cup \{24, 36\}, \\ T = {} & (S \setminus \{24\}) \cup \{48\}. \end{aligned}$$

Let b, c be the two cut indicators. Here $v(24) = 1$, $v(48) = -1$; the off-grid odd part has least period $e = 4$; and the two least mask periods are

$$p_0 = 4, \qquad p_1 = 12, \qquad p_2 = d = 24.$$

For example, $\mathcal{B}_4$ has the active residues $1 \pmod 4$ and the neutral residue $6 \pmod{12}$; its least period is twelve. The additional neutral classes $8, 16 \pmod{24}$ make $\mathcal{B}_{12}$ have least period twenty-four.

Choose $a = 24$. An initially valid residue witness is

$$t = 12, \qquad y_0 = 20, \qquad (r, s, z) = (11, 11, 6).$$

It has $\Sigma = 48$, while $W - t = 36$, so it does not fit. Here

$$W = 48, \qquad C_{\mathrm{box}} = 3 \cdot 12 + 24 = 60, \qquad W = C_{\mathrm{box}} - p_1.$$

The reflection is forced and gives

$$t' = 36, \qquad y_0' = 4, \qquad (r', s', z') = (1, 1, 6), \qquad \Sigma' = 12.$$

No padding is needed:

$$24 + 36 + 4 + 1 + 1 + 6 = 72.$$

For this positive list, the four factors in the detector are

$$v(24) = 1, \qquad b(60) - b(36) = -1, \qquad \mathcal{B}_{12}(40) - \mathcal{B}_{12}(4) = 1, \qquad H_4(1,1,6) = -2.$$

Thus its labeled discrepancy is -2. The only repeated numerical part is the pair of ones, so Lemma 2.1 gives ordinary discrepancy -1 at $(36, 24, 6, 4, 1, 1)$.

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Data availability. No datasets were generated or analyzed.

Computational verification. The proofs do not depend on numerical computation. Scripts used for verification are available at `https://github.com/johnnyshields/papers`.

Generative AI use. During the preparation of this work, the author used OpenAI's ChatGPT (GPT-6 Astra, GPT-6.1 Sol) and Anthropic's Claude (Opus 5.5, Sonnet 5.5) to assist with mathematical research, manuscript drafting, and the development of auxiliary verification code. After using these tools, the author reviewed and edited as needed. The author takes full responsibility for the content of the work.

References

[1] L. J. Billera, H. Thomas, and S. van Willigenburg, *Decomposable compositions, symmetric quasisymmetric functions and equality of ribbon Schur functions*, Adv. Math. **204** (2006), 204–240. doi:10.1016/j.aim.2005.05.014.

[2] P. R. W. McNamara and S. van Willigenburg, *Towards a combinatorial classification of skew Schur functions*, Trans. Amer. Math. Soc. **361** (2009), 4437–4470. doi:10.1090/S0002-9947-09-04683-2.

[3] V. Reiner, K. M. Shaw, and S. van Willigenburg, *Coincidences among skew Schur functions*, Adv. Math. **216** (2007), 118–152. doi:10.1016/j.aim.2007.05.006.